Chapter 4 : Point Set Topology

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Problem 4.1. If $\operatorname{card}(X) \geq 2$, there is a topology on X that is T_0 but not T_1.

Solution. For any distinct $x, y \in X$ consider the topology $\{\emptyset, \{x\}, \{x, y\}, X\}$, then we see that there is a open set containing x but not y but there are none containing y and not x so this is T_0 and not T_1. $\square$

Problem 4.2. If X is an infinite set, the cofinite topology on X is T_1 but not T_2, and is first countable iff X is countable.

Solution. If x, y are distinct elements of X then as $X \setminus \{x\}$ is open (it has a finite complenent $\{x\}$) we see that this is T_1 but this is not T_2 as we can not have two nonempty disjoint open sets to begin with, indeed if U, V are open and disjoint then we must have that $V \subseteq U^c$ which is finite ($U \neq \emptyset$ by assumption) and this in turn makes V^c infinite contradicting it being open in the first place as $V \neq \emptyset$ either.

If X is first countable then consider a countable base B_{x_0} of $x_0 \in X$. For $y \neq x$, $X \setminus \{y\}$ is a open neighbourhood of x and thus we can find $U \in B_{x_0}$ such that $x \in U \subset X \setminus \{y\}$ and thus, $y \in U^c$ with U^c being finite. From this we can write

$$X \setminus \{x_0\} = \bigcup_{U \in B_{x_0}} U^c$$

which is countable, being a countable union of countable (here, finite) sets so we are done. For the other direction, consider an ennumeration $\{x_n\}_{n \in \mathbb{N}}$, then we can construct a countable neighbourhood base of x_n as

$$\{X \setminus U : U \subseteq X \setminus \{x_n\}, U \text{ is finite}\}$$

infact, this contains exactly the open neighbourhoods of x_0 as these must be subsets of X which contain x_0 and have a finite complement, thus their complement has the form of U above and they are of form $X \setminus U$ as described. Its countable because the set of countable subsets of countable sets is countable. $\square$

Problem 4.3. Every metric space is normal. (If A, B are closed sets in the metric space (X, ρ), consider the sets of points where $\rho(x, A) < \rho(x, B)$ or $\rho(x, A) > \rho(x, B)$.)

Solution. We first prove that $x \mapsto \rho(x, A)$ is continuous when A is closed. For any $a \in A$ we see that $\rho(x, A) \leq \rho(x, a) \leq \rho(x, y) + \rho(y, a)$ and thus, $\inf_{a \in A} \rho(a, y) \geq \rho(x, A) - \rho(x, y)$, we can switch x, y and get another inequality, using both of these we find that $|\rho(x, A) - \rho(y, A)| \leq \rho(x, y)$ and thus this map is indeed continuous. Notice that for closed sets G we have that $G = \rho_G^{-1}(\{0\})$, this can be prove easily. Let, $\rho_A(x) := \rho(x, A)$, with this notation we see that if A, B are disjoint and closed then, the disjoint open sets $U = (\rho_A - \rho_B)^{-1}((0, \infty))$ and $V = (\rho_B - \rho_A)^{-1}((0, \infty))$ suffice as $x \in A$ implies $\rho_A(x) = 0$ while $\rho_B(x) > 0$ hence $A \subseteq V$ and similarly $B \subseteq U$. $\square$

Problem 4.4. Let $X = \mathbb{R}$ and let $\mathcal{T}$ be the family of all subsets of $\mathbb{R}$ of the form $U \cup (V \cap \mathbb{Q})$ where U, V are open in the usual sense. Then $\mathcal{T}$ is a topology that is Hausdorff but not regular. (In view of Exercise 3, this shows that a topology stronger than a normal topology need not be normal or even regular.)

Solution. This topology is stronger than the usual topology on $\mathbb{R}$ which is hausdorff and normal and thus this too is hausdorff and normal. Before we prove that it is not regular, it is worth to mention that intuitively speaking, regularity needn't carry on to stronger topologies as new closed sets might be introduced as well. Now, consider the closed set $\mathbb{Q}^c$ in this topology and let $x \in \mathbb{Q}$, then we see that if $U_1 \cup (V_1 \cap \mathbb{Q}) \supseteq \mathbb{Q}^c, U_2 \cup (V_2 \cap \mathbb{Q}) \ni x$ are open as described then we clearly need to have $U_1 \supseteq \mathbb{Q}^c$ and we claim that $U_1 \cap (V_2 \cup \mathbb{Q})$ is nonempty. We can find $a < b$ such that $x \in (a, b) \cap \mathbb{Q} \subseteq (V_2 \cap \mathbb{Q})$ as all the usual open sets in $\mathbb{R}$ are an union of open intervals. Now, we can cleary see that if $q \in U_1 \cap (a, b)$ then there is some $(c, d) \subseteq U_1 \cap (a, b)$ containing q, but this interval must have nonempty intersection with $\mathbb{Q} \cap (a, b)$ by the density of rationals, thus we can never find an open set containing x that is disjoint to some open set containing $\mathbb{Q}^c$ proving that $\mathcal{T}$ is not regular and thus not normal either (in this book, normal and regular spaces need to be T_1 by definition and thus the singletons are closed, making $T_3 \supset T_4$). $\square$

Problem 4.5. Every separable metric space is second countable.

Solution. For any countable dense subset S we have the following countable base $\{B(x, r) : r \in \mathbb{Q}, r > 0, x \in S\}$, the fact that this is a base can be easily proven. $\square$

Problem 4.6. Let $\mathcal{E} = \{(a, b] : -\infty < a < b < \infty\}$.

a. $\mathcal{E}$ is a base for a topology $\mathcal{T}$ on $\mathbb{R}$ in which the members of $\mathcal{E}$ are both open and closed.

b. $\mathcal{T}$ is first countable but not second countable. (If $x \in \mathbb{R}$, every neighborhood base at x contains a set whose supremum is x.)

c. $\mathbb{Q}$ is dense in $\mathbb{R}$ with respect to $\mathcal{T}$. (Thus the converse of Proposition 4.5 is false.)

Solution.

a. As $\mathcal{T}(\mathcal{E})$ contains all unions of elements in $\mathcal{E}$ we see that $(a, b]^c = (-\infty, a] \cup (b, \infty]$ is open in this topology as the rays can further be written as unions of the bounded h-open intervals from the base as well. This shows that the complement of each element of the basis is open so they themslves are closed.

b. For any $x \in \mathbb{R}$ we can consider $(x - 1, x] \ni x$. If $\mathcal{N}$ is any neighbourhood base thn there must exist some $U \in \mathcal{N}$ such that $x \in U \subseteq (x - 1, x]$ which trivially implies that $\sup U = x$ and thus $\mathcal{N}$ must be uncountable as $\mathbb{R}$ is so it is not second countable. To show that it is first countable it suffices to prove that for $x \in \mathbb{R}$ the following is a neighbourhood base : $\{(x - r, x] : r \in \mathbb{Q}, r > 0\}$. Now, gievn any $S \in \mathcal{T}(\mathcal{E})$ (due to proposition 4.4) is an union of finite intersections over $\mathcal{E}$ and thus if it contains x then we can find a finite intersection over $\mathcal{E}$, with x, that is a subset of S. Say its of form $\cap_{i=1}^n [x_i, x_i - \delta_i)$ then we see that the set $[x, x - \delta)$ for $\delta \in \mathbb{Q}$ such that $0 < \delta < \min_i \delta_i$ is contained in it and thus this is a neighbourhood base.

c. We prove the equivalent statement that $\emptyset = (\overline{\mathbb{Q}})^c = (\mathbb{Q}^c)^\circ$ so we have to prove that that there are no open sets in $\mathbb{Q}^c$. With knowledge of structure of open sets generated by some base it suffices to show that no finite intersections over this base are contained in $\mathbb{Q}^c$. We can use a fact used in chapter 1 that the intersection of finitely many h-intervals of this form is an union of some disjoint h-intervals so it suffices to show that no h-intervals are in $\mathbb{Q}^c$ which is trivially true by the density of $\mathbb{Q}$ in $\mathbb{R}$ in the usual sense. Thus, every seperable space is not second countable.

$\square$

Problem 4.7. If X is a topological space, a point $x \in X$ is called a cluster point of the sequence $\{x_j\}$ if for every neighborhood U of x, $x_j \in U$ for infinitely many j. If X is first countable, x is a cluster point of $\{x_j\}$ iff some subsequence of $\{x_j\}$ converges to x.

Solution. If some subsequence converges then the statement clearly holds, for the other direction first consider some neighbourhood base $\{U_n\}_{n\geq 0}$ and let $\{\cap_{m\leq n} U_m\}_{n\geq 0}$, we see that this is also a neighbourhood base and and V_n decreases. As infinitely many terms of the sequence are in any neighbourhood we can inductively choose $n_1 < n_2 < \dots$ such that $x_{n_k} \in V_k$. Now, we see that for any neighbourhood N of x we can find $V_k \subseteq N$ and thus, $x_{n_j} \in V_j \subseteq V_k \subseteq N$ for all $j \geq k$ so $x_{n_j} \to x$ by definition. $\square$

Problem 4.8. If X is an infinite set with the cofinite topology and $\{x_j\}$ is a sequence of distinct points in X, then $x_j \to x$ for every $x \in X$.

Solution. If $N \ni x$ is any neighbourhood then by definition N^c is finite and thus for large enough j we have $x_j \in N$ as the terms of the sequence are distinct, thus it must converge to x by definition. $\square$

Problem 4.9. If X is a linearly ordered set, the topology $\mathcal{T}$ generated by the sets $\{x : x < a\}$ and $\{x : x > a\}$ $(a \in X)$ is called the order topology.

a. If $a, b \in X$ and $a < b$, there exist $U, V \in \mathcal{T}$ with $a \in U$, $b \in V$, and $x < y$ for all $x \in U$ and $y \in V$. The order topology is the weakest topology with this property.

b. If $Y \subset X$, the order topology on Y is never stronger than, but may be weaker than, the relative topology on Y induced by the order topology on X.

c. The order topology on $\mathbb{R}$ is the usual topology.

Solution.

a. If there exists c such that $a < c < b$ then $U = \{x : x < c\}$ and $V = \{x : x > c\}$ work, otherwise let $U = \{x : x < b\}$ and $V = \{x : x > a\}$. If some topology J has this property then fix some $a \in X$, for all $x > a$ there exists U_x, V_x such that $a \in U_x, x \in V_x$ such that $u < v$ for all $u \in U, v \in V$ and thus we must have that $V_x \subseteq \{k : k > a\}$ and thus,
$$\bigcup_{X \ni x > a} V_x = \{k : k > a\} \in J$$
Similarly it can be shown that $\{k : k < a\} \in J$ so the order topology must be contained in it.

b. Its clearly never strictly stronger as the rays generating the order topology on Y must be in the base generating that on X as well. Inclusion can be strictly weaker however, for example consider $X = \mathbb{Z}$ with the usual order and let $Y = \{0, 1\}$ and we see that the order topology on this is $\mathcal{P}(Y) \subsetneq \mathcal{T}$

c. This follows from the fact that open sets in $\mathbb{R}$ in the usual sense are unions of open intervals and proposition 4.4.

$\square$

Problem 4.10. A topological space X is called disconnected if there exist nonempty open sets U, V such that $U \cap V = \emptyset$ and $U \cup V = X$; otherwise X is connected. When we speak of connected or disconnected subsets of X, we refer to the relative topology on them.

a. X is connected iff $\emptyset$ and X are the only subsets of X that are both open and closed.

b. If $\{E_\alpha\}_{\alpha \in A}$ is a collection of connected subsets of X such that $\bigcap_{\alpha \in A} E_\alpha \neq \emptyset$, then $\bigcup_{\alpha \in A} E_\alpha$ is connected.

c. If $A \subset X$ is connected, then $\overline{A}$ is connected.

d. Every point $x \in X$ is contained in a unique maximal connected subset of X, and this subset is closed. (It is called the connected component of x.)

Solution.

a. This follows directly from the definition.

b. For the sake of contradiction assume that $\cup E_\alpha$ is disconnected and the nonempty open sets U, V seperate it. Then, as E_α are all connected we either have $E_\alpha \subseteq U$ or $E_\alpha \subseteq V$ as otherwise $E_\alpha \cap U, E_\alpha \cap V$ is a valid seperation for E_α (in the relative topology of E_α.) But, as U, V are both nonempty we must have some $E_\alpha \subseteq U$ and $E_\beta \subseteq V$ which contradicts the fact that $\cap E_\alpha \neq \emptyset$.

c. Again for the sake of contradiction assume otherwise, let U, V be a seperation for $\overline{A}$. Then, if $A \cap U, A \cap V$ are both nonempty then this serves as a seperation in the relative topology for A and would imply that A was disconnected as well. We know that $\overline{A} = A \cup acc(A)$ and if $A \cap U$ was empty then we would have $U \subseteq acc(A)$ which implies that U has an accumulation point of A and thus by definition $U \cap A$ is nonempty which is a contradiction so both of these must be nonempty which is a contradiction, thus $\overline{A}$ must be connected to.

d. Let C be the union of all connected sets containing x. By part (b) we must have that C is connected too making C the maximal connected set containing x and as its closure is connected too by part (c) we must have that, $C = \overline{C}$ making it closed.

$\square$

Problem 4.11. If $E_1, \ldots, E_n$ are subsets of a topological space, the closure of $\bigcup_1^n E_j$ is $\bigcup_1^n \overline{E}_j$.

Solution. As $\cup E_j \subseteq \cup \overline{E_j}$ and finite unions of closed sets are closed we must have that $\overline{\cup E_j} \subseteq \cup \overline{E_j}$. Now, if C is a closed set containing $\cup E_j$ then it contains each E_j and thus their closures as well, thus $C \supseteq \cup \overline{E_j}$. Thus, by definition we have $\overline{\cup E_j} = \cap C \supseteq \cup \overline{E_j}$ where the intersection is over all closed sets containing $\cup E_j$. We thus have equality, having shown both sides of the set inclusion. $\square$

Problem 4.12. Let X be a set. A Kuratowski closure operator on X is a map $A \mapsto A^*$ from $\mathcal{P}(X)$ to itself satisfying (i) $\emptyset^* = \emptyset$, (ii) $A \subset A^*$ for all A, (iii) $(A^*)^* = A^*$ for all A, and (iv) $(A \cup B)^* = A^* \cup B^*$ for all A, B.

a. If X is a topological space, the map $A \mapsto \overline{A}$ is a Kuratowski closure operator. (Use Exercise 11.)

b. Conversely, given a Kuratowski closure operator, let $\mathcal{F} = \{A \subset X : A = A^*\}$ and $\mathcal{T} = \{U \subset X : U^c \in \mathcal{F}\}$. Then $\mathcal{T}$ is a topology, and for any set $A \subset X$, A^* is its closure with respect to $\mathcal{T}$.

Solution.

a. (iv) follows from Problem 4.11 and the rest are trivial.

b. As $\emptyset^* = \emptyset$ we see that $X \in \mathcal{T}$ and as $X \subseteq X^*$, we have $X = X^*$ so we similarly have $\emptyset \in \mathcal{T}$ too. Now, if $E_\alpha \in \mathcal{T}$ then we see that $E_\alpha^c = (E_\alpha^c)^*$ for all α and $\cup E_\alpha \in \mathcal{T}$ iff $\cap E_\alpha^c = (\cap E_\alpha^c)^*$. The $\subseteq$ direction follows from definition and as $\cap(E_\alpha)^c \subseteq E_\alpha^c = (E_\alpha^c)^*$, by (iv, which can be used to deduce monotonicity) we have $(\cap(E_\alpha)^c)^* \subseteq E_\alpha^c$ for all α and hence the other direction $(\cap E_\alpha^c)^* \subseteq \cap E_\alpha^c$ also holds. Closure under finite unions also follows from (iv) easily and we thus have that $\mathcal{T}$ is a topology. In this topology the closed sets are precisely the fixed points of this map and thus by monotonicty we see that,
$$A^* \subseteq \bigcap_{\substack{A \subseteq K \\ K \in \mathcal{F}}} K \subseteq A^*$$
as $A^* \in \mathcal{F}$ too, hence * is the closure operator with respect to $\mathcal{T}$.

□

Problem 4.13. If X is a topological space, U is open in X, and A is dense in X, then $\overline{U} = \overline{U \cap A}$.

Solution. It suffices to show that $U \subseteq \overline{U \cap A}$. Assume for the sake of contradiction that there exists some $x \in U$ such that $x \in (\overline{U \cap A})^c = H$ which is open. Thus, $U \cap H$ is a nonempty (they both contain x) open set and thus, as A is dense we must have that $A \cap (U \cap H)$ is nonempty which is absurd as $A \cap U \cap H = (A \cap U) \cap (U \setminus \overline{A \cap U}) = \emptyset$ so we are done. □

Problem 4.14. If X and Y are topological spaces, $f : X \to Y$ is continuous iff $f(\overline{A}) \subset \overline{f(A)}$ for all $A \subset X$ iff $\overline{f^{-1}(B)} \subset f^{-1}(\overline{B})$ for all $B \subset Y$.

Solution. If f is continuous then for all closed C containing $f(A)$ we must have that $A \subseteq f^{-1}(C)$ which is closed and thus $\overline{A} \subseteq f^{-1}(C)$ and thus, $f(\overline{A}) \subseteq f(f^{-1}(C)) \subseteq C$ so we must have that $f(\overline{A}) \subseteq \cap C = \overline{f(A)}$ where the union is over all such closed C. If the second hypothesis holds, then as
$$f^{-1}(\overline{B}) = \bigcap_{\substack{B \subseteq C \\ C \text{ closed}}} f^{-1}(C)$$
and $\overline{B} \subseteq C \implies f^{-1}(\overline{B}) \subseteq f^{-1}(C)$ for all the C as above, we see that the third hypothesis holds too. Now, if the third hypothesis holds then for all closed C we must

have $\overline{f^{-1}(C)} \subseteq f^{-1}(C)$ as $C = \overline{C}$ and this implies that $f^{-1}(C) = \overline{f^{-1}(C)}$ so preimages of closed sets are closed, which is equivalent to continuity so we are done. $\square$

Problem 4.15. If X is a topological space, $A \subset X$ is closed, and $g \in C(A)$ satisfies $g = 0$ on ∂A, then the extension of g to X defined by $g(x) = 0$ for $x \in A^c$ is continuous.

Solution. We see that $X = A \cup \overline{A^c}$ and as $\overline{A^c} = A^c \cup \partial A$, it must be the case that g is continuous on both $\overline{A^c}$ (its constant here) and A and hence, for any closed C

$$g^{-1}(C) = (g^{-1}(C) \cap A) \cup (g^{-1}(C) \cap \overline{A^c}) = g|_A^{-1}(C) \cup g|_{\overline{A^c}}^{-1}(C)$$

and as A and $\overline{A^c}$ are both closed we must have that $g|_A^{-1}(C)$ and $g|_{\overline{A^c}}^{-1}(C)$ are both closed. As finite unions of closed sets are closed, $g^{-1}(C)$ is closed so we are done. $\square$

Problem 4.16. Let X be a topological space, Y a Hausdorff space, and f, g continuous maps from X to Y.

a. $\{x : f(x) = g(x)\}$ is closed.

b. If $f = g$ on a dense subset of X, then $f = g$ on all of X.

Solution.

a. Consider $E = Y \times Y$ with the product topology and let $h : X \to E$ be defined by $h(x) := (f(x), g(x))$ which is continuous as its components are. Here, we see that,

$$\{x \in X : f(x) = g(x)\} = h^{-1}(\{(y, y) : y \in Y\})$$

and we claim that the diagonal Δ is closed. It suffices to prove that its complement is open, indeed if $x \neq y$ and U_x, U_y are disjoint open sets seperating them, we see that $U_x \times U_y$ is open in the product topology and

$$\Delta^c = \bigcup_{(x,y) \in \Delta^c} U_x \times U_y$$

is open so we are done.

b. This follows from the fact that closed dense sets are always the whole space itself.

$\square$

Problem 4.17. If X is a set, $\mathfrak{F}$ a collection of real-valued functions on X, and $\mathcal{T}$ the weak topology generated by $\mathfrak{F}$, then $\mathcal{T}$ is Hausdorff iff for every $x, y \in X$ with $x \neq y$ there exists $f \in \mathfrak{F}$ with $f(x) \neq f(y)$.

Solution. If x, y are distinct and there exists some f that maps them to different values then, as f is continuous over the weak topology, for disjoint open sets $U \ni f(x), V \ni f(y)$ we see that $f^{-1}(U) \ni x, f^{-1}(V) \ni y$ are disjoint open sets so $\mathcal{T}$ is hausdorff. Now, suppose this topology is hausdorff and we have $f(x) = f(y) = c$ for all f, for some distinct x, y. Now, for all open $U \in \mathbb{R}$ we see that $c \in U$ implies $f^{-1}(U)$ contains both x, y for all f, and otherwise it contains none of x, y for all f. Thus, all the sets in the generator either have both x, y or none of these and by proposition 4.4 the open sets too have this property and are unable to seperate these distinct points which makes $\mathcal{T}$ not hausdorff so we are done. $\square$

Problem 4.18. If X and Y are topological spaces and $y_0 \in Y$, then X is homeomorphic to $X \times \{y_0\}$ where the latter has the relative topology as a subset of $X \times Y$.

Solution. We can consider the bijection $x \mapsto (x, y_0)$. This is continuous as under the relative topology here, a generator of $X \times \{y_0\}$ is $\{U \times \{y_0\} : U \text{ open in } X\}$ and the preimages of these under this map are exactly the open sets in X, so we are done by proposition 4.9. $\square$

Problem 4.19. If $\{X_\alpha\}$ is a family of topological spaces, $X = \prod_\alpha X_\alpha$ (with the product topology) is uniquely determined up to homeomorphism by the following property: There exist continuous maps $\pi_\alpha : X \to X_\alpha$ such that if Y is any topological space and $f_\alpha \in C(Y, X_\alpha)$ for each α, there is a unique $F \in C(Y, X)$ such that $f_\alpha = \pi_\alpha \circ F$. (Thus X is the category-theoretic product of the X_α in the category of topological spaces.)

Solution. We describe the situation with a commutative diagram below.

$$\begin{tikzcd}
 & Y \arrow[dl, "f_\alpha"'] \arrow[d, "\exists! F"] \arrow[dr, "f_\beta"] & \\
X_\alpha & \prod X_\alpha \arrow[l, "\pi_\alpha"] \arrow[r, "\pi_\beta"'] & X_\beta
\end{tikzcd}$$

If $f_\alpha \in C(Y, X_\alpha)$ for all α then we see that $f_\alpha^{-1}(U_\alpha)$ is open in Y for all α, where U_α is open in X_α. If such a F were to exist then for this diagram to commute we must have that $\pi_\alpha \circ F = f_\alpha$ and thus, F is forced to be the map sending $y \mapsto (f_\alpha(y))_{\alpha \in A}$ and is also continiuous as all its components are due to how the product topology is defined and so it is indeed the unique such map in $C(Y, X)$. Now, suppose that Z had the same properties, let τ take the place of π and if $Y = X$ and $f_\alpha = \pi_\alpha \in C(X, X_\alpha)$ (the projections are continuous) then by definition there exists a map $G \in C(X, Z)$ such that $\pi_\alpha = \tau_\alpha \circ G$ i.e this diagram commutes

$$\begin{tikzcd}
 & \prod X_\alpha \arrow[dl, "\pi_\alpha"] \arrow[d, "G"'] \arrow[dr, "\pi_\beta"'] & \\
X_\alpha & Z \arrow[l, "\tau_\alpha"] \arrow[r, "\tau_\beta"'] & X_\beta
\end{tikzcd}$$

By definition there exists $H \in C(Z, X)$ such that $\tau_\alpha = \pi_\alpha \circ H$ implying, $\pi_\alpha = \pi_\alpha \circ (H \circ G)$ and similarly, $\tau_\alpha = \tau_\alpha \circ (G \circ H)$. Now, by uniqueness of F in the exercise statemen we must have that $H \circ G = e_X$ and $G \circ H = e_Z$ as $\pi_\alpha = \pi_\alpha \circ e_X$ and $\tau_\alpha = \tau_\alpha \circ e_Z$ which tells us that H is invertible and as composition of continuous maps are continuous, X and Z are homeomorphic through H so we are done. $\square$

Problem 4.20. If A is a countable set and X_α is a first (resp. second) countable space for each $\alpha \in A$, then $\prod_{\alpha \in A} X_\alpha$ is first (resp. second) countable.

Solution. Assume WLOG $A = \mathbb{N}$. If X_k are first countable then for some $x \in \prod X_k$ conisder the neighbourhood bases $\mathcal{N}(x_k)$ of x_k in X_k. Looking at the generators of the product topology of form $\cap_{j=1}^{n}\pi_{k_j}^{-1}(U_{k_j})$ containing x i.e. $U_{k_j} \ni x_{k_j}$ open in X_{k_j} we see that there exists $V_{k_j} \in \mathcal{N}(x_{k_j})$ such that $x_{k_j} \in V_{k_j} \subseteq U_{k_j}$ and thus,

$$x \in \bigcap_{j=1}^{n} \pi_{k_j}^{-1}(V_{k_j}) \subseteq \bigcap_{j=1}^{n} \pi_{k_j}^{-1}(U_{k_j})$$

so the following is a neighbourhood base for x in the product topology

$$\left\{ \bigcap_{j \in J} \pi_j^{-1}(N_j) : J \subseteq \mathbb{N}, J \text{ is finite}, (\forall j \in J) N_j \in \mathcal{N}(x_j) \right\}$$

We can construct an obvious injection from this to

$$\bigsqcup_{J \subseteq \mathbb{N}: J \text{ finite}} \underbrace{\prod_{j \in J} \mathbb{N}}_{\text{countable}}$$

which is countable, being a countable disjoint union of countable sets. Now, if they were second countable then the exact same construction gives us a base if we replace the countbale neighbourhood bases with bases. $\square$

Problem 4.21. If X is an infinite set with the cofinite topology, then every $f \in C(X)$ is constant.

Solution. By continuity we see that for any closed $C \subseteq \mathbb{R}$ we must have $f^{-1}(C)$ closed in X and hence finite, unless its all X. Now, suppose f is not constant and there are distinct $x, y \in f(X)$. Let, C_x, C_y be closed sets formed from $\mathbb{R}$ by removing small enough open intervals around x, y respectively such that $C_x \cup C_y = \mathbb{R}$. Then we see that $X = f^{-1}(\mathbb{R}) = f^{-1}(C_x) \cup f^{-1}(C_y)$ where both the preimages are closed and not the whole space as $x \neq y$ and thus they must be finite which is impossible as X is infinite. $\square$

Problem 4.22. Let X be a topological space, (Y, ρ) a complete metric space, and $\{f_n\}$ a sequence in Y^X such that $\sup_{x \in X} \rho(f_n(x), f_m(x)) \to 0$ as $m, n \to \infty$. There is a unique $f \in Y^X$ such that $\sup_{x \in X} \rho(f_n(x), f(x)) \to 0$ as $n \to \infty$. If each f_n is continuous, so is f.

Solution. Clearly for any fixed $x \in X$ the sequence $\{f_n(x)\}_n$ is cauchy and thus converges to say $f(x)$. For any $\epsilon > 0$ choose N such that $m, n \geq N$ implies $\rho(f_n(x), f_m(y)) < \epsilon$ for all $x \in X$ and thus $\rho(f(x), f_n(x)) \leq \rho(f_n(x), f_m(x)) + \rho(f_m(x), f(x)) < \epsilon + \rho(f_m(x), f(x))$ As, $m \to \infty$ we find that $\rho(f_m(x), f(x)) \to 0$ so we must have that $\rho(f_n(x), f(x)) \leq \epsilon$ and thus $\sup_{x \in X} \rho(f(x), f_n(x))$ must converge to 0 as $n \to \infty$. Now suppose that g was another such function, then as $\rho(f(x), g(x)) \leq \rho(f_n(x), f(x)) + \rho(g_n(x), g(x)) \to 0$ for all $x \in X$ we must have $f = g$ proving uniqueness. With the metric topology in mind it suffices to prove that for all $x \in X$ and $\epsilon > 0$ there exists a neighbourhood V of x such that $f(V) \subseteq B(f(x), \epsilon)$. First choose N such that $\rho(f(x), f_N(x)) < \epsilon/3$ for all x (we can do this by the previous part), then if V is a neighbournood of x such that $f_N(V) \subseteq B(f_N(x), \epsilon/3)$ then, for all $y \in V$

$$\rho(f(x), f(y)) \leq \rho(f_N(x), f_N(y)) + \rho(f_N(x), f(x)) + \rho(f_N(y), f(y)) < \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} = \epsilon$$

and thus $f(V) \subseteq B(f(x), \epsilon)$ and we are done as the balls around $f(x)$ form a neighbourhood base and this easily shows that f is continuous at all $x \in X$ and thus it is continuous over X. $\square$

Problem 4.23. Give an elementary proof of the Tietze extension theorem for the case $X = \mathbb{R}$.

Solution. Suppose $f \in C(A)$ with A closed. As open sets in $\mathbb{R}$ are disjoint unions of countably many open intervals. Say we have $A^c = \cup(a_n, b_n)$ where these intervals are disjoint; this might be a finite union and rays are also considered an interval. If (a_n, b_n) is a bounded interval then define F to be the line joining $f(a_n)$ to $f(b_n)$ on this, if its a ray (suppose left open wlog) $(-\infty, b)$ then set F to be $f(b)$ here and if $a \in A$ then set $F(a) = f(a)$. This definetly works but going through all the cases to prove this using sequential criterion is something I will postpone for later. □

Problem 4.24. A Hausdorff (it does not have to be) space X is normal iff X satisfies the conclusion of Urysohn's lemma iff X satisfies the conclusion of the Tietze extension theorem.

Solution. If X is normal then it satisisfies the conclusion of urysohn's lemma. If X satisfies the conclusion of ursohn's lemma then it satisfies the conclusion of tietze's extension theorem as normality of the space is not used in the proof, only tietze's theorem is used (look at Theorem 4.16) If X satisfies the conclusion of tietze's extension theorem we will prove that it is normal. Suppose A, B are disjoint closed sets and we have $f \in C(A \cup B, [0, 1])$ such that $f = 1/3$ on A and $f = 2/3$ on B and F be an extension then clearly $U = F^{-1}([0, 1/2)), V = F^{-1}((1/2, 1])$ are open and disjoint such that $A \subseteq U$ and $B \subseteq V$. □

Problem 4.25. If $(X, \mathcal{T})$ is completely regular, then $\mathcal{T}$ is the weak topology generated by $C(X)$.

Solution. If A is a closed set and $x \in A^c$ then we can find $f_x \in C(X)$ with $f_x(x) = 1$ and $f_x = 0$ on A. Now consider the closed sets $f_x^{-1}(\{0\})$, these contain A and exclude x, thus we see that,

$$A = \bigcap_{x \in A^c} f_x^{-1}(\{0\}) \quad \text{and} \quad A^c = \bigcup_{x \in A^c} f_x^{-1}(\mathbb{C} \setminus \{0\})$$

hence we see that all open sets (these can be expressed as the complement of a closed set all ways) are in the weak topology generated by $C(X)$ as this contains all sets of form $f^{-1}(\mathbb{C} \setminus \{0\})$ with $f \in C(X)$. Thus, $\mathcal{T}$ is contained in this weak topology (the inclusion in the other direction holds from the definition of weak topologies.) □

Problem 4.26. Let X and Y be topological spaces.

a. If X is connected (see Exercise 10) and $f \in C(X, Y)$, then $f(X)$ is connected.

b. X is called arcwise connected if for all $x_0, x_1 \in X$ there exists $f \in C([0, 1], X)$ with $f(0) = x_0$ and $f(1) = x_1$. Every arcwise connected space is connected.

c. Let $X = \{(s, t) \in \mathbb{R}^2 : t = \sin(s^{-1})\} \cup \{(0, 0)\}$, with the relative topology induced from $\mathbb{R}^2$. Then X is connected but not arcwise connected.

Solution.

a. If $f(X) = U \cup V$ with U, V nonempty, open and disjoint then we see that $X = f^{-1}(U) \cup f^{-1}(V)$ is also a valid seperation for X so if X is connected $f(X)$ must be connected too.

b. Suppose U, V are open, disjoint, nonempty, their union X and $x_0 \in U, x_1 \in V$. If $f \in C([0,1], X)$ is such that $f(0) = x_0$ and $f(1) = x_1$ then $f^{-1}(U), f^{-1}(V)$ are open, nonempty and partition $[0,1]$ which is a contradiction as $[0,1]$ is connected thus, arcwise connected spaces are always connected.

c. Let $G(x) = (x, \sin(1/x))$, then we see that $G(\mathbb{R}_+) \cup G(\mathbb{R}_-) \cup \{(0,0)\} = X$ and each of these are connected by part (a). It is easily provable that any open $U \ni (0,0)$ in the relative topology of X has nonempty intersection with $G(\mathbb{R}_+)$ and thus if V is another open set in the relative topology, disjoint to U such that $U \cup V = X$ then $G(\mathbb{R}_+) = (U \cap G(\mathbb{R}_+)) \cup (V \cap G(\mathbb{R}_+))$ is a valid seperation which is absurd as $G(\mathbb{R}_+)$ is connected so X must be connected. It is however not arc conected as we can not join $(0,0)$ to $(\pi, 0)$ through X with a continuous function. More rigourously, if $f : C([0,1], X)$ is such that $f(0) = (\pi, 0)$ and $f(1) = (0,0)$ then we would need $\|f(x)\|$ to be small enough, say less than $1/2$ for all $x \in (1-\delta, 1]$ for some small $\delta > 0$ so we must have that,

$$f((1-\delta, 1)) \subseteq J = \Big\{(s, \sin(1/s)) \in \mathbb{R}^2 : 1-\delta < s < 1, |\sin(1/s)| < 1/2\Big\}$$

But J is disconnected set with each connected component bounded away from $(0,0)$, i.e. they dont have this as an accumulation point. Thus, we see that $f((1-\delta, 1))$ must be in one of these components by part (a) and thus it does not have $f(1)$ as an accumulation point which is absurd when f is continuous.

□

Problem 4.27. If X_α is connected for each $\alpha \in A$ (see Exercise 10), then $X = \prod_{\alpha \in A} X_\alpha$ is connected. (Fix $x \in X$ and let Y be the connected component of x in X. Show that Y includes $\{y \in X : \pi_\alpha(y) = \pi_\alpha(x)$ for all but finitely many $\alpha\}$ and that the latter set is dense in X. Use Exercises 10 and 18.)

Solution. Let the set described in the question be S, then we find that

$$S = \bigcup_{\substack{J \subseteq A \\ J \text{ finite}}} \underbrace{\prod_{\alpha \in A} L_{\alpha, J}}_{\text{Let this be } P_J} \qquad \text{where } L_{\alpha,J} = X_\alpha \text{ if } \alpha \in J \text{ and } x_\alpha \text{ otherwise}$$

Now, for any $J = \{\alpha_1, \ldots, \alpha_n\}$ we can show using similar arguements as in Problem 1.18 that $\prod L_{\alpha,J}$ is homeomorphic to $X_{\alpha_1} \times \ldots \times X_{\alpha_n}$ which we show to be connected at the end, so its connectedd itslef. As all of these products contain x, thus Problem 4.10 implies that S is continuous so $S \subseteq Y$ by definition. If $\cap_1^n \pi_{\alpha_i}^{-1}(U_{\alpha_i})$ is any set in the conventional base for this space then we see that there is a point in this that only differs from x at the index $\alpha_1, ldots, \alpha_n$ so it has nonempty intersection with S and thus all open sets do too making S dense, thus we have $X = \overline{S} \subseteq \overline{Y} = Y$ as connected components are closed and we are done. To show that this is true for finite A as used as a fact earlier we can induct by adding a coordinate on every time in a sense. Say X, Y are connected then, the statement follows for $X \times Y$ directly using the proof above by Problem 18 as these individual spaces are connected. Now, we claim $(X \times Y) \times Z$

is homeomorphic to $X \times Y \times Z$ which will let us induct and prove it for finite A. Let, $f : (X \times Y) \times Z \to X \times Y \times Z$ be defined by $((x, y), z) \mapsto (x, y, z)$ then this is continuous, looking at $\pi_1 \circ f$ (the rest are proved to be continuous similarly) as, if U, V, W are open in X, Y, Z respectively then $(\pi_1 \circ f)^{-1}(U) = f^{-1}(U \times Y \times Z) = (U \times Y) \times Z$ is open in $(X \times Y) \times Z$ due to $U \times Y$ being open in $X \times Y$ and Z being open in Z, which is sufficient as components being continuous are equivalent to the function itself being continuous by the definition of the product topology. $\square$

Problem 4.28. Let X be a topological space equipped with an equivalence relation, $\tilde{X}$ the set of equivalence classes, $\pi : X \to \tilde{X}$ the map taking each $x \in X$ to its equivalence class, and $\mathcal{T} = \{U \subset \tilde{X} : \pi^{-1}(U) \text{ is open in } X\}$.

a. $\mathcal{T}$ is a topology on $\tilde{X}$. (It is called the quotient topology.)

b. If Y is a topological space, $f : \tilde{X} \to Y$ is continuous iff $f \circ \pi$ is continuous.

c. $\tilde{X}$ is T_1 iff every equivalence class is closed.

Solution.

a. Clearly $\emptyset, \tilde{X} \in \mathcal{T}$, the rest follows from π^{-1} commuting with unions and intersections.

b. Clearly π is continuous by definition and thus the only if direction is trivial, for the if direction it suffices to see that if $f \circ \pi$ is continuous then for all open $U \subseteq Y$, $\pi^{-1}(f^{-1}(U))$ is open, which implies that $f^{-1}(U)$ is open by definition so f must be continuous.

c. We use the equivalence that spaces are T_1 iff the singletons are closed. If $\{[x]\} \subseteq \tilde{X}$ were closed then $\pi^{-1}(\{[x]\}) = [x] \subseteq X$ is closed too as π is continuous and if the equivalence class are closed then for $[x] \neq [y]$ in $\tilde{X}$ we see that $[y]^c = \cup_{z \notin [y]}[z]$ is open in X and omits y while containing x and $\cup_{z \notin [y]}\{[z]\}$ is open in $\tilde{X}$ (its preimage under π is $[y]^c$ which is open in X) and omits $[y]$ while containing $[x]$ so the quotient topology for this must be T_1.

$\square$

Problem 4.29. If X is a topological space and G is a group of homeomorphisms from X to itself, G induces an equivalence relation on X, namely, $x \sim y$ iff $y = g(x)$ for some $g \in G$. Let $X = \mathbb{R}^2$; describe the quotient space $\tilde{X}$ and the quotient topology on it (as in Exercise 28) for each of the following groups of invertible linear maps. In particular, show that in (a) the quotient space is homeomorphic to $[0, \infty)$; in (b) it is T_1 but not Hausdorff; in (c) it is T_0 but not T_1, and in (d) it is not T_0. (In fact, in (d) X is uncountable, but there are only six open sets and there are points $p \in \tilde{X}$ such that $\overline{\{p\}} = \tilde{X}$.)

a. $\left\{\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} : \theta \in \mathbb{R}\right\}$

b. $\left\{\begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix} : a \in \mathbb{R}\right\}$

c. $\left\{\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a > 0, b \in \mathbb{R}\right\}$

d. $\left\{ \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} : a, b \in \mathbb{Q} \setminus \{0\} \right\}$

Solution.

a. These are rotation matirces, each equivalence class consists of vectors of a specific length. Let $f : \tilde{X} \to [0, \infty)$ be defined by $[x] \mapsto r$ where $\|x\| = r \geq 0$, its well defined by the previous line. Using Problem 4.28 we see that this is continuous as $f \circ \pi$ is the norm i.e. $f \circ \pi : x \mapsto \|x\|$, which is continuous and thus so is f. It is clearly a bijection, hence $\tilde{X}$ is homeomorphic to $[0, \infty)$.

b. These are shear matrices and if we let the one associated to $a \in \mathbb{R}$ be g_a then we see that $g_a(x, y) = (x + ay, y)$ from this we see that the equivalence classes are $[(x, 0)] = \{(x, 0)\}$ and $[(x, y)] = \mathbb{R} \times \{y\}$ for $y \neq 0$. By Problem 28, its T_1 as the equivalence classes are all closed in $\mathbb{R}^2$ but its clearly not haussdorff as we can not find any open sets seperating the $[(1, 0)]$ and $[(2, 0)]$ for example : if U is open and $[(1, 0)] \in U$ then $\pi^{-1}(U)$ is also open and contains $(1, 0)$ which forces it to contain $(1, y)$ for all y close enough to 0 (but not 0) and hence $[(1, y)] \in \pi(\pi^{-1}(U)) \subseteq U$ and if V was an open set containing $[(2, 0)]$ then it too would contain all the equivalence classes $[(2, y)] = [(1, y)]$ for small enough nonzero y and thus $U \cap V \neq \emptyset$, so $\tilde{X}$ is not hausdorff.

c. Here the equivalence classes are $[(0, 0)] = \{(0, 0)\}, [(x, 0)] = \mathbb{R}_+ \times \{0\}$ if $x > 0$ and $\mathbb{R}_- \times \{0\}$ if $x < 0$, $[(x, y)] = \mathbb{R} \times \{y\}$ if $y \neq 0$. Now, consider the points $[(0, 0)]$ and $[(1, 0)]$, if U is any open set contianing $[(0, 0)]$ then by logic similar to before, it always contains $[(1, 0)]$ but we the open set $V = \tilde{X} \setminus \{[(0, 0)]\}$ (it is open as $\pi^{-1}(U) = \mathbb{R}^2 \setminus \{(0, 0)\}$) does not contain $[(0, 0)]$ while containing $[(1, 0)]$ so its not T_1. From From the above we see that $[(0, 0)]$ can be seperated from any point so now assume that none of the two chosen classes are $[(0, 0)]$. If exactly one of them has points with y coordinates being 0 then either one of the open sets $U = \tilde{X} \setminus \{[(0, 0)], [(1, 0)]\}$ or $V = \tilde{X} \setminus \{[(0, 0)], [(-1, 0)]\}$ do the job ($\pi^{-1}(U) = \mathbb{R}^2 \setminus \{(x, 0) : x \leq 0\}$ and $\pi^{-1}(V) = \mathbb{R}^2 \setminus \{(x, 0) : x \geq 0\}$.) Now, if both the points are equivalence classes with nonzero y coordinates then they can be seperated by open sets which correspond to open tubes in $\mathbb{R}^2$ i.e. sets of form $\mathbb{R} \times (a, b)$ so all together, this is a T_0 space.

d. Let $Q = \mathbb{Q} \setminus \{0\}$ for brevity and let $A = \sqrt{2}Q \times \sqrt{2}Q$ and $B = Q \times Q$ be two different equivalence classes, we show that they can not be seperated even in the T_0 sense. If U is open and contains $A \in U$ then as $A \subseteq \pi^{-1}(U)$ we find that there is an open ball in $\mathbb{R}^2$ around the point $(\sqrt{2}, \sqrt{2})$ that is a subset of $\pi^{-1}(U)$ and this clearly contains a point from B so we must have that $B \in \pi(\pi^{-1}(U))$ and the exact same arguement relying on the fact that aQ is dense in $\mathbb{R}$ for any nonzero a, shows that no open set containing B can exclude A so the space is not T_0. For the extra questions at the end : it is uncountable because each equivalence class is countable whereas the real plane is uncountable, the six open sets are (here $\mathbb{R} \setminus \{0\}$ is written as R) :

$$\tilde{X}, \emptyset, \pi(\mathbb{R}^2 \setminus \{(0, 0)\}), \pi(\mathbb{R} \times R), \pi(R \times \mathbb{R}), \pi(R \times R),$$

Now, a point p for which $\{p\}$ is dense in $\tilde{X}$ can be found from the above to be $p = Q \times Q$ as it has nonempty intersection with every nonempty open set in $\tilde{X}$.

□

Problem 4.30. If A is a directed set, a subset B of A is called cofinal in A if for each $\alpha \in A$ there exists $\beta \in B$ such that $\beta \geq \alpha$.

a. If B is cofinal in A and $\langle x_\alpha \rangle_{\alpha \in A}$ is a net, the inclusion map $B \to A$ makes $\langle x_\beta \rangle_{\beta \in B}$ a subnet of $\langle x_\alpha \rangle_{\alpha \in A}$.

b. If $\langle x_\alpha \rangle_{\alpha \in A}$ is a net in a topological space, then $\langle x_\alpha \rangle$ converges to x iff for every cofinal $B \subset A$ there is a cofinal $C \subset B$ such that $\langle x_\gamma \rangle_{\gamma \in C}$ converges to x.

Solution.

□

Problem 4.31. Let $\langle x_n \rangle_{n \in \mathbb{N}}$ be a sequence.

a. If $k \mapsto n_k$ is a map from $\mathbb{N}$ to itself, then $\langle x_{n_k} \rangle_{k \in \mathbb{N}}$ is a subnet of $\langle x_n \rangle$ iff $n_k \to \infty$ as $k \to \infty$, and it is a subsequence (as defined in §0.1) iff n_k is strictly increasing in k.

b. There is a natural one-to-one correspondence between the subsequences of $\langle x_n \rangle$ and the subnets of $\langle x_n \rangle$ defined by cofinal sets as in Exercise 30.

Solution.

□

Problem 4.32. A topological space X is Hausdorff iff every net in X converges to at most one point. (If X is not Hausdorff, let x and y be distinct points with no disjoint neighborhoods, and consider the directed set $\mathcal{N}_x \times \mathcal{N}_y$ where $\mathcal{N}_x, \mathcal{N}_y$ are the families of neighborhoods of x, y.)

Solution.

□

Problem 4.33. Let $\langle x_\alpha \rangle_{\alpha \in A}$ be a net in a topological space, and for each $\alpha \in A$ let $E_\alpha = \{x_\beta : \beta \geq \alpha\}$. Then x is a cluster point of $\langle x_\alpha \rangle$ iff $x \in \bigcap_{\alpha \in A} \overline{E}_\alpha$.

Solution.

□

Problem 4.34. If X has the weak topology generated by a family $\mathfrak{F}$ of functions, then $\langle x_\alpha \rangle$ converges to $x \in X$ iff $\langle f(x_\alpha) \rangle$ converges to $f(x)$ for all $f \in \mathfrak{F}$. (In particular, if $X = \prod_{i \in I} X_i$, then $x_\alpha \to x$ iff $\pi_i(x_\alpha) \to \pi_i(x)$ for all $i \in I$..)

Solution.

□

Problem 4.35. Let X be a set and $\mathcal{A}$ the collection of all finite subsets of X, directed by inclusion. Let $f : X \to \mathbb{R}$ be an arbitrary function, and for $A \in \mathcal{A}$, let $z_A = \sum_{x \in A} f(x)$. Then the net $\langle z_A \rangle$ converges in $\mathbb{R}$ iff $\{x : f(x) \neq 0\}$ is a countable set $\{x_n\}_{n \in \mathbb{N}}$ and $\sum_1^\infty |f(x_n)| < \infty$, in which case $z_A \to \sum_1^\infty f(x_n)$. (Cf. Proposition 0.20.)

Solution.

□

Problem 4.36. Let X be the set of Lebesgue measurable complex-valued functions on $[0,1]$. There is no topology $\mathcal{T}$ on X such that a sequence $\langle f_n \rangle$ converges to f with respect to $\mathcal{T}$ iff $f_n \to f$ a.e. (Use Corollary 2.32 and Exercises 30b and 31b.)

Solution.

□

Problem 4.37. Let $0'$ denote a point that is not an element of $(-1,1)$, and let $X = (-1,1) \cup \{0'\}$. Let $\mathcal{T}$ be the topology on X generated by the sets $(-1,a)$, $(a,1)$, $[(-1,b) \setminus \{0\}] \cup \{0'\}$, and $[(c,1) \setminus \{0\}] \cup \{0'\}$ where $-1 < a < 1$, $0 < b < 1$, and $-1 < c < 0$. (One should picture X as $(-1,1)$ with the point 0 split in two.)

a. Define $f, g : (-1,1) \to X$ by $f(x) = x$ for all x, $g(x) = x$ for $x \neq 0$, and $g(0) = 0'$. Then f and g are homeomorphisms onto their ranges.

b. X is T_1 but not Hausdorff, although each point of X has a neighborhood that is homeomorphic to $(-1,1)$ (and hence is Hausdorff).

c. The sets $[-\frac{1}{2}, \frac{1}{2}]$ and $([-\frac{1}{2}, \frac{1}{2}] \setminus \{0\}) \cup \{0'\}$ are compact but not closed in X, and their intersection is not compact.

Solution.

□

Problem 4.38. Suppose that $(X, \mathcal{T})$ is a compact Hausdorff space and $\mathcal{T}'$ is another topology on X. If $\mathcal{T}'$ is strictly stronger than $\mathcal{T}$, then $(X, \mathcal{T}')$ is Hausdorff but not compact. If $\mathcal{T}'$ is strictly weaker than $\mathcal{T}$, then $(X, \mathcal{T}')$ is compact but not Hausdorff.

Solution.

□

Problem 4.39. Every sequentially compact space is countably compact.

Solution.

□

Problem 4.40. If X is countably compact, then every sequence in X has a cluster point. If X is also first countable, then X is sequentially compact.

Solution.

□

Problem 4.41. A T_1 space X is countably compact iff every infinite subset of X has an accumulation point.

Solution.

□

Problem 4.42. The set of countable ordinals (§0.4) with the order topology (Exercise 9) is sequentially compact and first countable but not compact. (To prove sequential compactness, use Proposition 0.19.)

Solution.

$\square$

Problem 4.43. For $x \in [0, 1)$, let $\sum a_n(x)2^{-n}$ ($a_n(x) = 0$ or 1) be the base-2 decimal expansion of x. (If x is a dyadic rational, choose the expansion such that $a_n(x) = 0$ for n large.) Then the sequence $\langle a_n \rangle$ in $\{0, 1\}^{[0,1)}$ has no pointwise convergent subsequence. (Hence $\{0, 1\}^{[0,1)}$, with the product topology arising from the discrete topology on $\{0, 1\}$, is not sequentially compact. It is, however, compact, as we shall show in §4.6.)

Solution.

$\square$

Problem 4.44. If X is countably compact and $f : X \to Y$ is continuous, then $f(X)$ is countably compact.

Solution.

$\square$

Problem 4.45. If X is normal, then X is countably compact iff $C(X) = BC(X)$. (Use Exercises 40 and 44. If $\langle x_n \rangle$ is a sequence in X with no cluster point, then $\{x_n : n \in \mathbb{N}\}$ is closed, and Corollary 4.17 applies.)

Solution.

$\square$

Problem 4.46. Prove Theorem 4.34.

Solution.

$\square$

Problem 4.47. Prove Proposition 4.36. Also, show that if X is Hausdorff but not locally compact, Proposition 4.36 remains valid except that X^* is not Hausdorff.

Solution.

$\square$

Problem 4.48. Complete the proof of Proposition 4.40.

Solution.

$\square$

Problem 4.49. Let X be a compact Hausdorff space and $E \subset X$.

a. If E is open, then E is locally compact in the relative topology.

b. If E is dense in X and locally compact in the relative topology, then E is open. (Use Exercise 13.)

c. E is locally compact in the relative topology iff E is relatively open in $\overline{E}$.

Solution.

$\square$

Problem 4.50. Let U be an open subset of a compact Hausdorff space X and U^* its one-point compactification (see Exercise 49a). If $\phi : X \to U^*$ is defined by $\phi(x) = x$ if $x \in U$ and $\phi(x) = \infty$ if $x \in U^c$, then ϕ is continuous.

Solution.

$\square$

Problem 4.51. If X and Y are topological spaces, $\phi \in C(X, Y)$ is called proper if $\phi^{-1}(K)$ is compact in X for every compact $K \subset Y$. Suppose that X and Y are LCH spaces and X^* and Y^* are their one-point compactifications. If $\phi \in C(X, Y)$, then ϕ is proper iff ϕ extends continuously to a map from X^* to Y^* by setting $\phi(\infty_X) = \infty_Y$.

Solution.

$\square$

Problem 4.52. The one-point compactification of $\mathbb{R}^n$ is homeomorphic to the n-sphere $\{x \in \mathbb{R}^{n+1} : |x| = 1\}$.

Solution.

$\square$

Problem 4.53. Lemma 4.37 remains true if the assumption that X is locally compact is replaced by the assumption that X is first countable.

Solution.

$\square$

Problem 4.54. Let $\mathbb{Q}$ have the relative topology induced from $\mathbb{R}$.

a. $\mathbb{Q}$ is not locally compact.

b. $\mathbb{Q}$ is σ-compact (it is a countable union of singleton sets), but uniform convergence on singletons (i.e., pointwise convergence) does not imply uniform convergence on compact subsets of $\mathbb{Q}$.

Solution.

$\square$

Problem 4.55. Every open set in a second countable LCH space is σ-compact.

Solution.

$\square$

Problem 4.56. Define $\Phi : [0, \infty] \to [0, 1]$ by $\Phi(t) = \frac{t}{t+1}$ for $t \in [0, \infty)$ and $\Phi(\infty) = 1$.

a. Φ is strictly increasing and $\Phi(t + s) \leq \Phi(t) + \Phi(s)$.

b. If (Y, ρ) is a metric space, then $\Phi \circ \rho$ is a bounded metric on Y that defines the same topology as ρ.

c. If X is a topological space, the function $\rho(f, g) = \Phi(\sup_{x \in X} |f(x) - g(x)|)$ is a metric on $\mathbb{C}^X$ whose associated topology is the topology of uniform convergence.

d. If X is a σ-compact LCH space and $\{U_n\}_1^\infty$ is as in Proposition 4.39, the function

$$\rho(f, g) = \sum_1^\infty 2^{-n} \Phi(\sup_{x \in U_n} |f(x) - g(x)|)$$

is a metric on $\mathbb{C}^X$ whose associated topology is the topology of uniform convergence on compact sets.

Solution.

□

Problem 4.57. An open cover $\mathcal{U}$ of a topological space X is called locally finite if each $x \in X$ has a neighborhood that intersects only finitely many members of $\mathcal{U}$. If $\mathcal{U}, \mathcal{V}$ are open covers of X, $\mathcal{V}$ is a refinement of $\mathcal{U}$ if for each $V \in \mathcal{V}$ there exists $U \in \mathcal{U}$ with $V \subset U$. X is called paracompact if every open cover of X has a locally finite refinement.

a. If X is a σ-compact LCH space, then X is paracompact. In fact, every open cover $\mathcal{U}$ has locally finite refinements $\{V_\alpha\}$, $\{W_\alpha\}$ such that $\overline{V}_\alpha$ is compact and $W_\alpha \subset V_\alpha$ for all α. (Let $\{U_n\}_1^\infty$ be as in Proposition 4.39. For each n, $\{E \cap (U_{n+2} \setminus U_{n-1}) : E \in \mathcal{U}\}$ is an open cover of $\overline{U}_{n+1} \setminus U_n$. Choose a finite subcover to obtain $\{V_\alpha\}$ and mimic the beginning of the proof of Proposition 4.41 to obtain $\{W_\alpha\}$.)

b. If X is a σ-compact LCH space, for any open cover $\mathcal{U}$ of X there is a partition of unity on X subordinate to $\mathcal{U}$ and consisting of compactly supported functions.

Solution.

□

Problem 4.58. If $\{X_\alpha\}_{\alpha \in A}$ is a family of topological spaces of which infinitely many are noncompact, then every closed compact subset of $\prod_{\alpha \in A} X_\alpha$ is nowhere dense.

Solution.

□

Problem 4.59. The product of finitely many locally compact spaces is locally compact.

Solution.

□

Problem 4.60. The product of countably many sequentially compact spaces is sequentially compact. (Use the "diagonal trick" as in the proof of Theorem 4.44.)

Solution.

□

Problem 4.61. Theorem 4.43 remains valid for maps from a compact Hausdorff space X into a complete metric space Y provided the hypothesis of pointwise boundedness is replaced by pointwise total boundedness. (Make this statement precise and then prove it.)

Solution.

□

Problem 4.62. Rephrase Theorem 4.44 in a form similar to Theorem 4.43 by using the metric in Exercise 56d.

Solution.

□

Problem 4.63. Let $K \in C([0,1]\times[0,1])$. For $f \in C([0,1])$, let $Tf(x) = \int_0^1 K(x,y)f(y)\,dy$. Then $Tf \in C([0,1])$, and $\{Tf : \|f\|_u \leq 1\}$ is precompact in $C([0,1])$.

Solution.

□

Problem 4.64. Let (X,ρ) be a metric space. A function $f \in C(X)$ is called Hölder continuous of exponent α ($\alpha > 0$) if the quantity

$$N_\alpha(f) = \ldots \sup_{x\neq y} \frac{|f(x)-f(y)|}{\rho(x,y)^\alpha}$$

is finite. If X is compact, $\{f \in C(X) : \|f\|_u \leq 1 \text{ and } N_\alpha(f) \leq 1\}$ is compact in $C(X)$.

Solution.

□

Problem 4.65. Let U be an open subset of $\mathbb{C}$, and let $\{f_n\}$ be a sequence of holomorphic functions on U. If $\{f_n\}$ is uniformly bounded on compact subsets of U, there is a subsequence that converges uniformly to a holomorphic function on compact subsets of U. (Use the Cauchy integral formula to obtain equicontinuity.)

Solution.

□

Problem 4.66. Let $1 - \sum_1^\infty c_n t^n$ be the Maclaurin series for $(1-t)^{1/2}$.

a. The series converges absolutely and uniformly on compact subsets of $(-1,1)$, as does the termwise differentiated series $-\sum_1^\infty nc_n t^{n-1}$. Thus, if $f(t) = 1 - \sum_1^\infty c_n t^n$ then $f'(t) = -\sum_1^\infty nc_n t^{n-1}$.

b. By explicit calculation, $f(t) = -2(1-t)f'(t)$, from which it follows that $(1-t)^{-1/2}f(t)$ is constant. Since $f(0) = 1$, $f(t) = (1-t)^{1/2}$.

Solution.

□

Problem 4.67. Prove Theorem 4.52. (If there exists $x_0 \in X$ such that $f(x_0) = 0$ for all $f \in \mathcal{A}$, let Y be the one-point compactification of $X \setminus \{x_0\}$; otherwise let Y be the one-point compactification of X. Apply Proposition 4.36 and the Stone-Weierstrass theorem on Y.)

Solution.

□

Problem 4.68. Let X and Y be compact Hausdorff spaces. The algebra generated by functions of the form $f(x, y) = g(x)h(y)$, where $g \in C(X)$ and $h \in C(Y)$, is dense in $C(X \times Y)$.

Solution.

□

Problem 4.69. Let A be a nonempty set, and let $X = [0, 1]^A$. The algebra generated by the coordinate maps $\pi_\alpha : X \to [0, 1]$ ($\alpha \in A$) and the constant function 1 is dense in $C(X)$.

Solution.

□

Problem 4.70. Let X be a compact Hausdorff space. An ideal in $C(X, \mathbb{R})$ is a subalgebra I of $C(X, \mathbb{R})$ such that if $f \in I$ and $g \in C(X, \mathbb{R})$ then $fg \in I$.

a. If I is an ideal in $C(X, \mathbb{R})$, let $h(I) = \{x \in X : f(x) = 0 \text{ for all } f \in I\}$. Then $h(I)$ is a closed subset of X, called the hull of I.

b. If $E \subset X$, let $k(E) = \{f \in C(X, \mathbb{R}) : f(x) = 0 \text{ for all } x \in E\}$. Then $k(E)$ is a closed ideal in $C(X, \mathbb{R})$, called the kernel of E.

c. If $E \subset X$, then $h(k(E)) = \overline{E}$.

d. If I is an ideal in $C(X, \mathbb{R})$, then $k(h(I)) = \overline{I}$. (Hint: $k(h(I))$ may be identified with a subalgebra of $C_0(U, \mathbb{R})$ where $U = X \setminus h(I)$.)

e. The closed subsets of X are in one-to-one correspondence with the closed ideals of $C(X, \mathbb{R})$.

Solution.

□

Problem 4.71. (This is a variation on the theme of Exercise 70; it does not use the Stone-Weierstrass theorem.) Let X be a compact Hausdorff space, and let M be the set of all nonzero algebra homomorphisms from $C(X, \mathbb{R})$ to $\mathbb{R}$. Each $x \in X$ defines an element $\hat{x}$ of M by $\hat{x}(f) = f(x)$.

a. If $\phi \in M$ then $\{f \in C(X, \mathbb{R}) : \phi(f) = 0\}$ is a maximal proper ideal in $C(X, \mathbb{R})$.

b. If I is a proper ideal in $C(X, \mathbb{R})$, there exists $x_0 \in X$ such that $f(x_0) = 0$ for all $f \in I$. (Suppose not; construct an $f \in I$ with $f > 0$ everywhere and conclude that $1 \in I$. This requires no deep theorems.)

c. The map $x \to \hat{x}$ is a bijection from X to M.

d. If M is equipped with the topology of pointwise convergence, then the map $x \to \hat{x}$ is a homeomorphism from X to M. (Since M is defined purely algebraically, it follows that the topological structure of X is completely determined by the algebraic structure of $C(X, \mathbb{R})$.)

Solution.

□

Problem 4.72. Every subset of a completely regular space is completely regular in the relative topology.

Solution.

□

Problem 4.73. If X is a completely regular space, a subalgebra $\mathcal{A}$ of $BC(X)$ is called completely regular if (i) it is closed and contains the constant functions, and (ii) $\mathcal{A} \cap C(X, I)$ separates points and closed sets.

a. If (Y, e) is a Hausdorff compactification of X, $\mathcal{A}_Y = \{f \circ e : f \in C(Y)\}$ is a completely regular subalgebra of $BC(X)$.

b. If (Y, e) and (Y', e') are Hausdorff compactifications of X such that $\mathcal{A}_Y = \mathcal{A}_{Y'}$, there is a homeomorphism $\phi : Y \to Y'$ such that $\phi \circ e = e'$. (Adapt the proof of Theorem 4.57, which deals with the case $Y = \beta X$.)

c. If (Y, e) is the compactification of X associated to $\mathfrak{F} \subset C(X, I)$, then $\mathcal{A}_Y$ is the smallest closed subalgebra of $BC(X)$ that contains $\mathfrak{F}$. (Use Exercise 69.)

d. The Hausdorff compactifications of X are in one-to-one correspondence with the completely regular subalgebras of $BC(X)$.

Solution.

□

Problem 4.74. Consider $\mathbb{N}$ (with the discrete topology) as a subset of its Stone-Čech compactification $\beta\mathbb{N}$.

a. If A and B are disjoint subsets of $\mathbb{N}$, their closures in $\beta\mathbb{N}$ are disjoint. (Hint: $\chi_A \in C(\mathbb{N}, I)$.)

b. No sequence in $\mathbb{N}$ converges in $\beta\mathbb{N}$ unless it is eventually constant (so $\beta\mathbb{N}$ is emphatically not sequentially compact).

Solution.

□

Problem 4.75. Suppose X is a completely regular space. The set M of nonzero algebra homomorphisms from $BC(X, \mathbb{R})$ to $\mathbb{R}$, equipped with the topology of pointwise convergence, is homeomorphic to βX. (See Exercise 71. This realization of βX is the natural one from the point of view of Banach algebra theory.)

Solution.

$\square$

Problem 4.76. If X is normal and second countable, there is a countable family $\mathfrak{F} \subset C(X, I)$ that separates points and closed sets. (Let $\mathcal{B}$ be a countable base for the topology. Consider the set of pairs $(U, V) \in \mathcal{B} \times \mathcal{B}$ such that $\overline{U} \subset V$ and use Urysohn's lemma.)

Solution.

$\square$

Problem 4.77. Let $\{(X_n, \rho_n)\}_1^\infty$ be a countable family of metric spaces whose metrics take values in $[0, 1]$. (The latter restriction can always be satisfied; see Exercise 56b.) Let $X = \prod_{n=1}^\infty X_n$. If $x, y \in X$, say $x = (x_1, x_2, \dots)$ and $y = (y_1, y_2, \dots)$, define $\rho(x, y) = \sum_1^\infty 2^{-n} \rho_n(x_n, y_n)$. Then ρ is a metric that defines the product topology on X.

Solution.

$\square$